Code No: 136EV **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
DATA STRUCTURES
(Common to EEE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is the need of asymptotic notation?	[2]
b)	Compare data type with data structure.	[3]
c)	In which situations a stack is preferred over a queue?	[2]
d)	What are the challenges in implementing circular queue?	[3]
e)	What is a max heap? Give an example.	[2]
f)	List the applications of threaded binary tree.	[3]
g)	What are the limitations of binary search?	[2]
h)	How can a heap be used in sorting?	[3]
i)	What is a trie? Give example.	[2]
j)	What is the significance of colours in red-black tree?	[3]

PART – B

(50 Marks)

2.a) How to compute best case complexity of an algorithm? Give illustration.
b) Write an algorithm to find largest of three given numbers and perform its analysis. [5+5]

OR

3. Write an algorithm to concatenate two single linked list and explain it by tracing with appropriate input. [10]

4.a) Demonstrate usage of stack in converting an infix expression into a postfix expression.
b) How to implement queue using stacks? Give your justification. [5+5]

OR

5. Write a C program to create a double ended queue using arrays and perform insertion and deletion operations. [10]

6.a) Explain the properties of binary tree.
b) Write a program to implement max priority queue. [5+5]

OR

7.a) Which graph representation approach is preferred in programming? Justify your answer.
b) Make a comparison of breadth first search and depth first search traversals. [5+5]

8.a) How to handle collisions in hashing? Explain with an example.
b) Compare the algorithm complexities of linear search and binary search. [5+5]

OR

9. Sort the following numbers using radix sort, explain each pass. [10]
3423 1238 9017 7756 5487 9836 2567 1100

10. With suitable example, demonstrate insertion and deletion operations on a binary search tree. [10]

OR

11.a) Illustrate different types of rotations used for AVL trees.
b) Write the steps of Knuth-Morris-Pratt algorithm for pattern matching. [5+5]

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